

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

- First , convert 1 e V to Joules : $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$
- Use the equation $E = (hc) / \lambda$ where E is energy , h is Planck's constant , c is the speed of light , and λ is wavelength .
- Rearrange to solve for wavelength , $\lambda = (hc) / E$
- Plug in the values . $h \approx 6.626 \times 10^{-34} \text{ Js}$, $c \approx 3 \times 10^8 \text{ m/s}$, $E \approx 1.602 \times 10^{-19} \text{ J}$
- Result is $\lambda \approx 1.23 \times 10^{-10} \text{ m}$
- Convert meters to cm by multiplying by 100
- Therefore , $\lambda \approx 1.23 \times 10^{-8} \text{ cm}$
- Close st match in answer choices is G.

</think>

<answer>

G

</answer>